

Himanshu Kumar

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Professional Summary

Computer Science student at IIIT Delhi specializing in mobile applications and AI-driven solutions. **Designed and deployed** AI-powered mobile application detecting plant diseases with 96% accuracy using ResNet-50 and PyTorch Mobile. **Directed** multiple collaborative projects in agri-tech and game development, delivering scalable software. Strong foundation in algorithms, system design, full-stack development, Android app creation, and edge-optimized deep learning models.

Technical Skills

Programming Languages:	Python, Java, Kotlin, C++, C, JavaScript, SQL, HTML/CSS
Mobile Development:	Android Development, Android Studio, Jetpack Compose, CameraX, Android Navigation, Material Design, Mobile UI/UX
Machine Learning & AI:	PyTorch, PyTorch Mobile, TensorFlow, ResNet-50, CNN, Deep Learning, Computer Vision, OpenAI SDK, Llama-2, NLP
Databases & Backend:	MySQL, SQLite, Firebase, REST APIs, Room Database, Backend Development
Development Tools:	Git, GitHub, VS Code, IntelliJ IDEA, Linux, Docker, Agile/Scrum, CI/CD
Core Competencies:	Data Structures & Algorithms, System Design, OOP, Software Architecture, Full-Stack Development, Problem Solving

Projects

Plant Disease Detection Mobile Application

Under Prof. Arani Bhattacharya — Mobile Computing *Individual — Jan 2024 – May 2024*

- **Created** Android app identifying 38+ plant diseases with ResNet-50
- **Optimized** PyTorch Mobile offline inference for rural low-connectivity regions
- **Integrated** Llama-2-based advisory system for contextual treatment guidance
- **Crafted** responsive UI using Jetpack Compose with live camera pipeline

Technologies: Kotlin, PyTorch Mobile, Jetpack Compose, CameraX
GitHub Repository

Interactive Stick Hero Game

Under Prof. Raghava Mutharaju — Advanced Programming *Team of 2 — Nov 2023 – Dec 2023*

- **Built** 2D game with physics simulation and responsive controls using JavaFX
- **Applied** MVC and OOP principles for scalable architecture
- **Tested** gameplay mechanics to ensure smooth user experience

Technologies: Java, JavaFX, OOP, MVC
GitHub Repository

Zoo Management System

Object-Oriented Programming Project *Individual — Oct 2023 – Nov 2023*

- **Implemented** modular management system using Factory, Strategy, and Observer patterns
- **Developed** CLI with validation, persistence, and maintainable architecture

Technologies: Java, OOP, Design Patterns

Sobha Limited Financial Analysis (FY 2021–2025)
Foundation of Finance Course Project *Team of 3 — Oct 2025*

- **Analyzed** financial performance including revenue, margins, liquidity, and efficiency ratios
- **Conducted** competitive research, CSR assessment, and strategic evaluation
- **Prepared** cash flow studies, common-size statements, SWOT analysis, and SIP simulations
- **Delivered** comprehensive report with actionable recommendations

Technologies: Financial Modeling, Excel, Ratio Analysis, Market Research
GitHub Repository

Professional Experience

Software Developer & Research Assistant
IIIT Delhi – Plant Disease Detection Research *Jan 2024 – May 2024*

- **Optimized** mobile-ready deep learning pipeline reducing inference time by 40%
- **Collaborated** with agricultural experts to ensure user-centric design
- **Presented** research outcomes to faculty panel, receiving recognition

Software Engineering Team Lead
IIIT Delhi – Academic Projects *Aug 2022 – Present*

- **Directed** 2–4 member teams to deliver 5+ projects on schedule
- **Enhanced** team workflow efficiency by 25% via mentorship and structured code reviews
- **Managed** project planning, testing, documentation, and agile coordination

Education

IIIT Delhi *2022 – 2026 (Expected)*
B.Tech in Computer Science and Engineering CGPA: 6.33/10

RSBV Patparganj, Delhi *2019 – 2020*
Class XII (Science) 84%

RSBV School Block, Shakarpur *2017 – 2018*
Class X 82%

Relevant Coursework: DS Algo, OS, DBMS, ML, Mobile Computing, SE Principles, System Design, Computer Networks

Achievements & Recognition

- **Achieved** 96% accuracy with AI model for agricultural applications
- **Delivered** 5+ software projects on schedule with high-quality outcomes
- **Boosted** team productivity by 25% via leadership and mentorship
- **Recognized** by IIIT Delhi faculty for innovation in mobile computing
- **Contributed** actively to open-source repositories

Professional Interests

- **Emerging Tech:** AI/ML, mobile development, agri-tech solutions
- **Sports:** Table Tennis, Cricket, teamwork and strategy
- **Tech Trends:** Computer vision, mobile computing, sustainable tech